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Barcode reading engine

M1V2

Features

ÿ Proprietary core technologies

Employing core decoding technology with independent intellectual property rights, it possesses excellent... 1D/2D decoding capability.

Excellent screen coding capabilities

Special calibration was performed for screen barcodes to adapt to large data volume screen barcodes with low brightness and various screen protectors.

Multiple interfaces

It provides USB and TTL232 interfaces to meet more application needs.

High integration

The image acquisition unit and decoding board are integrated into one unit, offering high integration and adaptability to various product integration applications.

Application scenarios

Parcel lockers, ticket gates, display kiosks, and various self-service application devices, etc.

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M1V2/Barcode Reading Engine

Scanning performance	sensor		640*480 CMOS
	illumination		White LED
	Focus		Red LED
	Code recognition system	2D	QR Code, Micro QR, Data Matrix, PDF417, Micro PDF417, Aztec, Maxicode, Hanxin Code, Dotcode, Composite
		1D	Code 128 (ISBT 128, AIM 128, GS1 128), EAN-13, EAN-8, UPC-E, UPC-A, ISBN, ISSN, Codell, Intereaved 2 of 5, Code39, Code93, Code32, Codabar, Matrix 2of5, Industrial25 IATA25, MSI Plessey, GS 1 DataBar, Plessey, Febraban, Composite, etc.
	Reading accuracy*		ÿ3mil
	Typical depth of field reading*	EAN-13	50mm-330mm (13mil 13 bytes)
		Code 39	50mm-120mm (5mil 7 bytes)
		Code128	50mm-340mm (13mil 10 bytes)
		QR Code	30mm-215mm (15mil 30 bytes)
		Data Matrix	40mm-145mm (10mil 100 bytes)
		PDF 417	55mm-140mm (6.67mil 30 bytes)
	Symbol contrast*		ÿ10%
	Scanning angle**		360° turning angle, ±55° elevation angle, ±55° yaw angle
	Field of view		Horizontal 43°, Vertical 32°
Mechanical/Electrical Parameter	Communication Interface		TTL - Serial port, USB (HID-KBW, virtual serial port)
	External dimensions (mm)		21.5mm*8.5mm*7.1mm
	Operating voltage/current		DC3.3V/160mA
	Standby current		11mA (USB) 22mA (UART)
Environmental parameters	operating temperature		-20ÿ~+55ÿ
	Storage temperature		-40ÿ~+70ÿ
	Operating humidity		5%~95% (non-condensing)
	Ambient lighting		0~100,000 LUX
Certification			CE, FCC, RoHS, optical safety (IEC62471)

Test conditions: Ambient temperature = 23°C; Ambient illuminance = 360 LUX (fluorescent lamp); Customized test code used for paper code.

*Deflection angle test conditions: Test distance = (minimum depth + maximum depth) / 2; 2D: QR V2:

*Specifications are subject to change without notice.

*V1.1 Release

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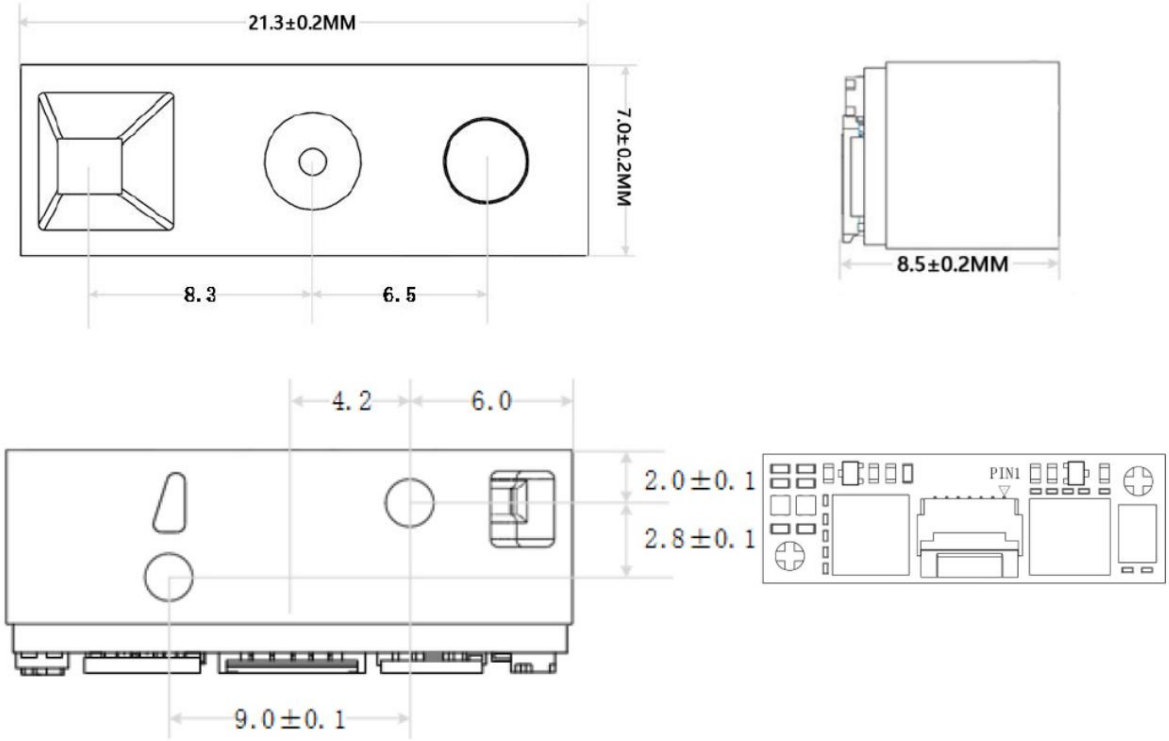
Dimensions

When integrating the M1V2, please refer to the physical dimensions shown in the diagram below (default tolerance: ±0.5mm). In structural design, ensure that other components do not compress the M1V2 devices.

(Unit: mm)



There are two M1.4 screw holes on the bottom for installation. The image below shows the correct placement or installation when the screw holes are facing down. (Note: Screws) (Installation depth not exceeding 2mm).

The specific signal definitions for the 13-PIN FPC are as follows:

PIN	definition	I/O	illustrate
1	GND	-	negative power input
2	nTRIG	I	The trigger key input signal, held low for more than 5ms, will trigger code reading.
3	nRST	I	Reset signal input, active low. Holding low for more than 100μs will reset the device.
4	led	O	The indicator light outputs a signal; when idle, it outputs a low level.
5	BUZZER	O	The passive buzzer outputs a low-level signal when idle.
6	EXT.LED.CTRL	O	External lighting control signal
7	USB_D+	-	USB_D+ signal
8	USB_D-	-	USB_D- signal
9	UART_TX	O	TTL level transmission
10	UART_RX	I	TTL level reception
11	GND	I	negative power input
12	VCC	I	Power input positive 3.3V
13	VCC	I	Power input positive 3.3V

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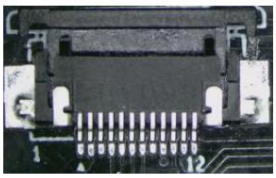
To ensure compatibility with other commonly available 12-pin interfaces, it is recommended that customers use 12-pin bottom-contact connectors (0.5mm pitch) in their designs. This is primarily for user equipment.

When using the device, it is recommended to use the following wiring sequence for the 12-pin lower contact connector interface (the definitions of TTL serial ports and I/O ports are based on the user's device as the host, therefore the serial port's receiving and transmitting...).

(The input/output definitions of the I/O ports are exactly the opposite of those of M1V2).

Pin number	Signal name	I/O type	Default state	Function Description
1	NC	-	-	-
2	VDD	-	-	3.3V power input
3	GND	-	-	power ground
4	TXD	Output	-	TTL level 232 serial port signal transmission
5	RXD	enter	-	TTL level 232 serial signal reception
6	USB_D-	-	-	USB_D-signal
7	USB_D+	-	-	USB_D+ signal
8	NC	-	-	-
9	BUZ	enter	-	Buzzer input signal, high effective
10	led	enter	-	Indicator light input signal, high active value
11	nRST	Output	-	Reset output signal, active low
12	nTRIG	Output	-	Trigger output signal, active low

The recommended 12-pin lower contact connector is shown in the image below, from left to right: Pin 1-Pin 12.